

Random Number Generation and Monte Carlo Simulation

EEC 3150

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1 RNG

- Random Walks
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RNG

Random Walks

Random Walks:

A random walk is a sequence $x_0, x_1, x_2, \dots, x_{N-1}$ of randomly generated steps through some space:

- It is usually formulated

$$x_{i+1} = x_i + r_i \quad (1)$$

where $i = 0, \dots, N - 1$, x_0 is a given initial state, and the r_i 's are randomly generated from some distribution

Create some random walks:

- 1D $+1$ or -1
- 1D $\text{uniform}(0,1)$
- 1D $\text{uniform}(-1,1)$
- 1D normal distribution
- 2D normal
- 2D up-down/left-right (with bias)

RNG on Different Domains:

We can use NumPy's RNG tools to randomly generate points inside an arbitrary domain:

- We can uniformly sample different domains by sampling along the “dimensions” of the domain

Uniformly sample some domains:

- Rectangle
- Circle perimeter
- Disk area
- Sphere

Monte Carlo Simulation

Monte Carlo Simulation

Monte Carlo Simulation:

Monte Carlo simulation is a computational technique used to model and simulate complex systems or processes using random sampling and statistical methods

You can use it to:

- Evaluate statistics
- Sample complex distributions
- Evaluate complex integrals over complex domains

Technique:

- Generate LARGE volume of random samples on some distribution
- Perform various statistical analyses of the result

Monte Carlo Simulation

Estimate π :

Steps:

1. Uniformly sample N_{sq} points in the unit square
2. Inscribe the unit circle in the unit square
3. Count the number N_{cl} points in the unit circle
4. For large N_{sq} , $\frac{N_{cl}}{N_{sq}} \approx \frac{A_{cl}}{A_{sq}}$
5. $\frac{A_{cl}}{A_{sq}} = \frac{\pi r^2}{(2r)^2} = \frac{\pi}{4} \approx \frac{N_{cl}}{N_{sq}}$
6. Therefore, $\pi \approx 4 \frac{N_{cl}}{N_{sq}}$

Some notes:

- More accurate approximations for large N_{sq}
- MC has a slow rate of convergence, i.e., it takes MANY samples to get accurate results (there are more efficient means of estimating π)

Monte Carlo Simulation

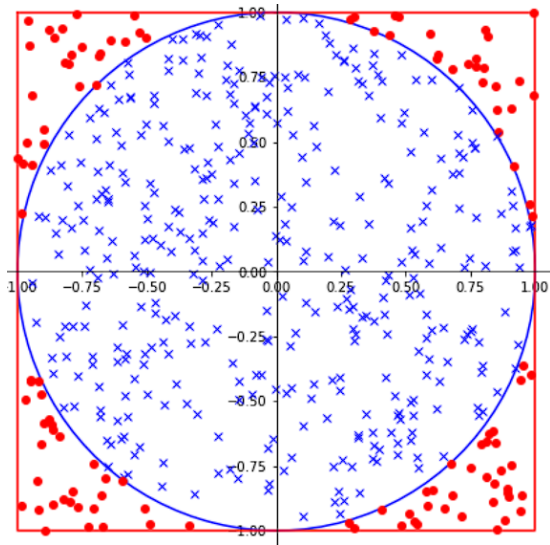


Figure: A circle inscribed in a square with randomly generated points.

Monte Carlo Simulation

Integrate $f(x)$:

Assume $f(x) > 0$ for all $x \in [a, b]$

Evaluate $\int_a^b f(x)dx$

1. Uniformly sample N_{rect} points on the rectangle $[a, b] \times [0, M]$ where $M \geq \max(f(x))$ over $[a, b]$
2. For large N_{rect} , $\frac{N_{u.c.}}{N_{rect}} \approx \frac{A_{u.c.}}{A_{rect}}$ (u.c. = under the curve)
3. Therefore, $A_{u.c.} \approx A_{rect} \frac{N_{u.c.}}{N_{rect}}$

Some notes:

- More accurate approximations for large N_{rect} , slow rate of convergence
- More points required for higher dimensional integrals
- Can be used to integrate where $f(x) < 0$ but you must account for positive and negative values separately

Monte Carlo Simulation

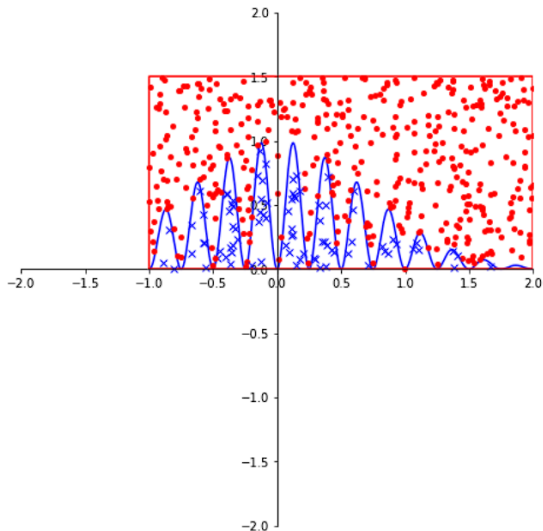


Figure: Integrating a function with Monte Carlo.

Monte Carlo Simulation

Calculate statistics of a complex distribution:

A cannon can fire a cannonball with an initial velocity v_i at an angle θ . However, the cannon is imperfect, thus, there is a small range of error for each of these values.

Using the projectile motion equations

$$x_f = x_i + v_i \cos(\theta)t$$

$$y_f = y_i + v_i \sin(\theta)t - \frac{1}{2}gt^2$$

find the distribution of flight times t and firing range x_f given that

- $v_i = 10 \pm 0.5$ m/s (uniformly distributed)
- $\theta = 45 \pm 3$ degrees (uniformly distributed)
- $x_f, y_f = 0$ m
- $g = 9.8$ m/s²